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Senior Secondary School Term II Examination, 2022

Marking Scheme – PHYSICS (SUBJECT CODE – 042)

(PAPER CODE – 55/4/1)

General Instructions: -

1. You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully.
2. **“Evaluation policy is a confidential policy as it is related to the confidentiality of the examinations conducted, Evaluation done and several other aspects. Its’ leakage to public in any manner could lead to derailment of the examination system and affect the life and future of millions of candidates. Sharing this policy/document to anyone, publishing in any magazine and printing in News Paper/Website etc may invite action under IPC.”**
3. Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one’s own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. **However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and marks be awarded to them. In class-X, while evaluating two competency based questions, please try to understand given answer and even if reply is not from marking scheme but correct competency is enumerated by the candidate, marks should be awarded.**
4. The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
5. Evaluators will mark($\sqrt{\quad}$) wherever answer is correct. For wrong answer ‘X’ be marked. Evaluators will not put right kind of mark while evaluating which gives an impression that answer is correct and no marks are awarded. **This is most common mistake which evaluators are committing.**
6. If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled. This may be followed strictly.
7. If a question does not have any parts, marks must be awarded in the left-hand margin and encircled. This may also be followed strictly.
8. If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out.
9. No marks to be deducted for the cumulative effect of an error. It should be penalized only once.

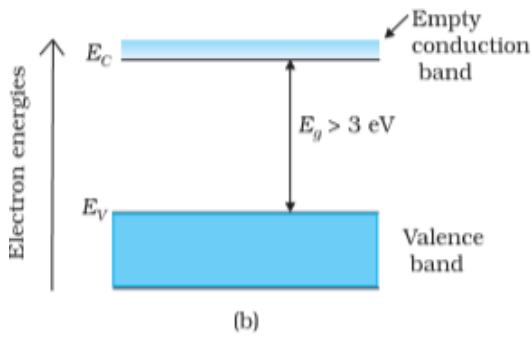
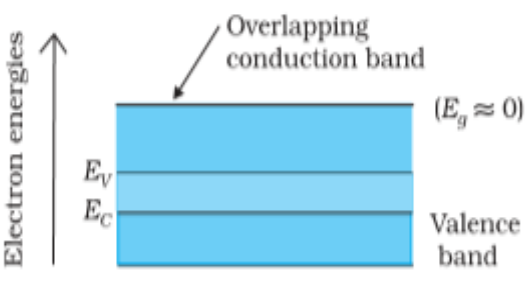
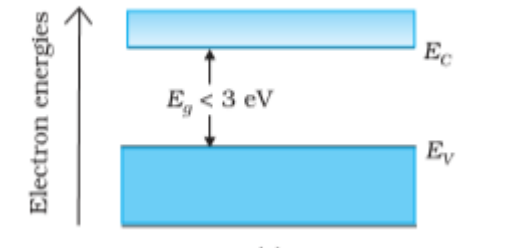
10. A full scale of marks 0-35 has to be used. Please do not hesitate to award full marks if the answer deserves it.
11. Every examiner has to necessarily do evaluation work for full working hours i.e. 8 hours every day and evaluate 30 answer books per day in main subjects and 35 answer books per day in other subjects (Details are given in Spot Guidelines). This is in view of the reduced syllabus and number of questions in question paper.
12. Ensure that you do not make the following common types of errors committed by the Examiner in the past:-
 - Leaving answer or part thereof unassessed in an answer book.
 - Giving more marks for an answer than assigned to it.
 - Wrong totaling of marks awarded on a reply.
 - Wrong transfer of marks from the inside pages of the answer book to the title page.
 - Wrong question wise totaling on the title page.
 - Wrong totaling of marks of the two columns on the title page.
 - Wrong grand total.
 - Marks in words and figures not tallying.
 - Wrong transfer of marks from the answer book to online award list.
 - Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.)
 - Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
13. While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as cross (X) and awarded zero (0) Marks.
14. Any unassessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
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MARKING SCHEME

Senior Secondary School Examination TERM–II, 2022

PHYSICS (Subject Code–042)

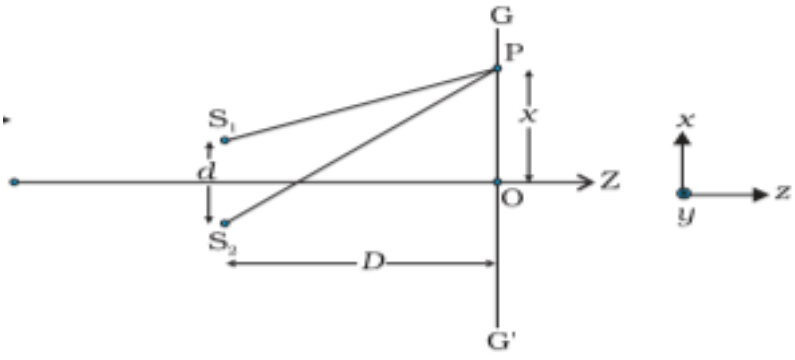
[Paper Code : 55/4/1]

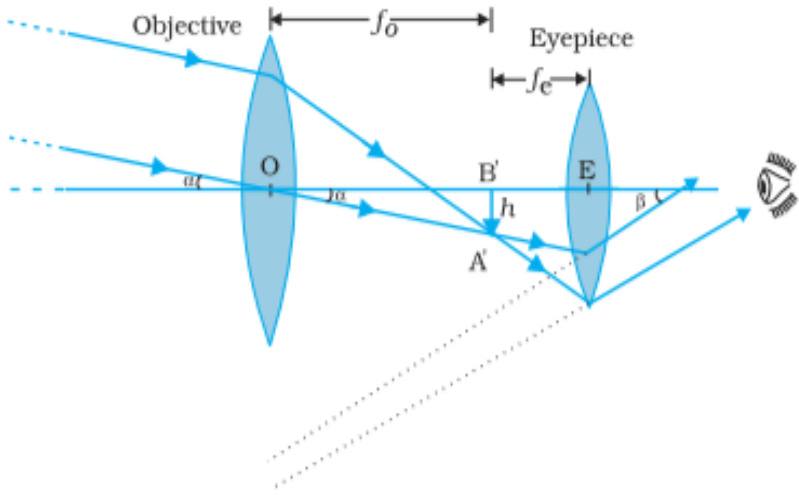
Q. No.	EXPECTED ANSWER / VALUE POINTS	Marks	Total Marks
1.	- Meaning of energy Band gap $\frac{1}{2}$ - Energy Band Diagram of Conductor, Insulator & Semiconductor $\frac{1}{2}$ $(\frac{1}{2} + \frac{1}{2} + \frac{1}{2})$ The gap between the top of the valence band and bottom of the conduction band is called the energy band gap.  (b) ENERGY BAND DIAGRAM FOR INSULATOR  ENERGY BAND DIAGRAM FOR CONDUCTOR  ENERGY BAND DIAGRAM FOR SEMICONDUCTOR	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	2

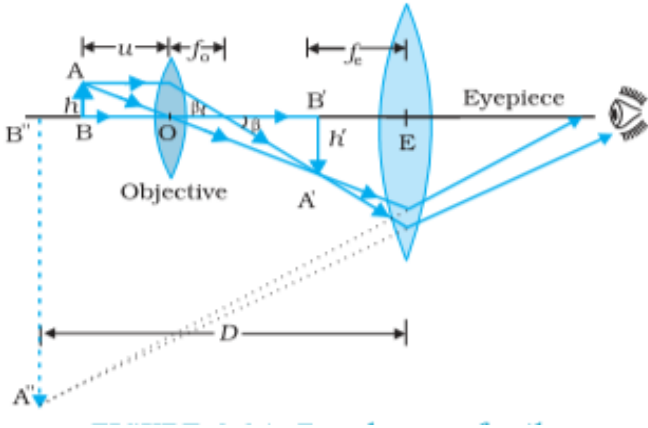
2.	• Name of Spectral Series ½ • Ratio of the wavelength 1½ a) Balmer Series $\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$ $\frac{1}{\lambda_{\max}} = R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) \quad (n_f = 2, \quad n_i = 3)$ $\lambda_{\max} = \frac{36}{5R}$ $\frac{1}{\lambda_{\min}} = R \left(\frac{1}{2^2} - \frac{1}{\infty} \right) \quad (n_f = 2, \quad n_i = \infty)$ $\lambda_{\min} = \frac{4}{R}$ $\frac{\lambda_{\max}}{\lambda_{\min}} = \frac{9}{5}$ OR • Meaning of matter waves ½ • Ratio of de- Broglie wavelength 1½ b) Wave associated with moving material particle $\lambda = \frac{h}{\sqrt{2qmV}}$ $\therefore \frac{\lambda_P}{\lambda_\alpha} = \sqrt{\frac{q_\alpha m_\alpha}{q_P m_P}}$ $\therefore \frac{\lambda_P}{\lambda_\alpha} = \sqrt{\frac{2e \cdot 4m}{e \cdot m}}$ $= \frac{2\sqrt{2}}{1}$	½ ½ ½ ½ ½ ½ ½ ½	2
3.	• Naming the device ½ • Any three Uses 1½		

	Light emitting diode (LED) Advantages of LED (Any three of the following) (i) Low operational voltage and less power (ii) Fast on-off switching capability (iii) The bandwidth of emitted light is 100 Å to 500 Å or in other words it is nearly (but not exactly) monochromatic. (iv) Long life and ruggedness. (v) Fast action and no warm up time required (Note- Give credit of half mark to the students for writing electric bulb or other relevant device, and further credit 1½ marks if they mention the advantages of written device only)	½ ½+½+½	2
	SECTION-B		
4.	a) Distinguishing nuclear fission & fusion 1 b) Calculation of duration 2 a) Nuclear fission – The process of breaking a very heavy nucleus into lighter nuclei, having mass number in the range of middle mass number ($30 < A < 170$). Nuclear Fusion- It is the process of joining of very light nuclei ($A \leq 10$) to form a heavier nucleus. b) No. of atoms in 100 g $= \frac{6.023 \times 10^{23}}{2} \times 100 = 3.0115 \times 10^{25}$ Energy released/atom $= \frac{3.27 \text{ MeV}}{2} = 1.635 \text{ MeV}$ Total energy released $= 3.0115 \times 10^{25} \times 1.635 \text{ MeV}$ $= 3.0115 \times 1.635 \times 10^{25} \times 1.6 \times 10^{-13}$ $= 7.878 \times 10^{12} \text{ J}$ $t = \frac{E}{P}$ $= \frac{7.878 \times 10^{12} \text{ J}}{500 \text{ J/s}} = 1.5756 \times 10^{10} \text{ s}$ $= \frac{1.5756 \times 10^{10}}{315 \times 10^7} \simeq 500 \text{ y}$	½ ½ ½ ½ ½ ½ ½	3

	Alternatively : $E = \frac{MQ}{2m_d}$ $= \frac{(0.1 \text{ kg}) \times (3.27 \text{ MeV})}{2(2.04) \times (1.66 \times 10^{-27} \text{ kg} / 4)}$ $= 0.0492 \times 10^{27} = 4.92 \times 10^{25} \text{ MeV}$ $t = \frac{E}{P}$ $= \frac{(4.92 \times 10^{25}) \times (1.6 \times 10^{-13})}{500} = 1.5756 \times 10^{10} \text{ s}$ $= \frac{1.5756 \times 10^{10}}{3.15 \times 10^7} \simeq 500y$	1/2 1/2 1/2 1/2	
5.	a) For explaining reason1 b) For explaining reason1 c) For explaining reason1 a) When a <i>p-n</i> junction is forward biased, the junction width decreases and as a result, its resistance also decreases. On the other hand, when a <i>p-n</i> junction is reverse biased, the junction width increases hence resistance increases. b) Conductivity of intrinsic semi-conductors is very low. Hence, no electronic device can be developed using them. Dopping increases conductivity, hence makes intrinsic semiconductor suitable for making electronic devices. c) It is easier to observe the change in the current with change in light intensity if a reverse bias is applied. Alternatively The fractional change due to photo-effects on the minority charge carriers dominated reverse bias current, is more easily measurable than the fractional change in forward bias current.	1/2 1/2 1 1	3
6.	a) Calculation of distance of closest approach1½ b) Calculation of distance of closest approach1½		

	a) For α particles, distance of closest approach $r_{\alpha} = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{E_k}$ $r_{\alpha} = \frac{9 \times 10^9 \times 2 \times 79 \times (1.6 \times 10^{-19})^2}{2.56 \times 10^{-12}}$ $= 14.22 \times 10^{-15} \text{ m}$ $= 14.22 \text{ fm}$ b) For proton, distance of closest approach $r_p = \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{E_k}$ $r_p = \frac{r_{\alpha}}{2}$ $= 7.11 \times 10^{-15} \text{ m}$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	3
7.	• Explanation of bright and dark fringes. 1 • Derivation of the expression of fringe width. 2 We get bright fringes, when waves from two coherent sources meet at a point on the screen with a phase difference $\Delta\phi = 2n\pi \quad (n = 1, 2, 3, \dots)$ We get dark fringes, when waves from two coherent sources meet at a point on the screen with a phase difference $\Delta\phi = (2n+1)\pi \quad (n = 1, 2, 3, \dots)$ (Note: Give full credit for explaining using path difference)  For maxima, $S_2P - S_1P = n\lambda \quad n = 0, 1, 2, 3, \dots$ (1)	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	

	From figure, $(S_2P)^2 - (S_1P)^2 = \left[D^2 + \left(x + \frac{d}{2} \right)^2 \right] - \left[D^2 + \left(x - \frac{d}{2} \right)^2 \right] = 2xd$ $S_2P - S_1P = \frac{2xd}{S_2P + S_1P}$ $S_2P \approx S_1P \approx D$ $S_2P - S_1P = \frac{2xd}{2D} = \frac{xd}{D} \quad (2)$ From equation (1) & (2) $n\lambda = \frac{xd}{D}$ $x_n = \frac{n\lambda D}{d} \quad \text{for } n^{\text{th}} \text{ maxima}$ Similarly for $(n+1)^{\text{th}}$ maxima $x_{n+1} = \frac{(n+1)\lambda D}{d}$ $\therefore \text{fringe width} = x_{n+1} - x_n = \frac{\lambda D}{d}$ Note- Give full credit, if a student derives this relation using condition of minimum intensity.	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	3
8.	(a) (i) Labelled ray diagram of astronomical telescope— 1 (ii) (I) Calculation of length of tube 1 (II) Calculation of magnification 1 a) (i) 	1	

	(ii) Given $f_o = 150 \text{ cm}$, $f_e = 6 \text{ cm}$ (I) Length of the tube $L = f_o + f_e$ $= 150 + 6$ $L = 156 \text{ cm}$ (II) $m = \frac{f_o}{f_e}$ $\frac{150}{6} = 25$ OR b) (i) Labelled ray diagram of compound microscope – 1 (ii) (I) Position of image calculation 1 (II) Calculation of linear magnification 1 b) i)  Ray diagram of image formation by a compound microscope ii) Given $u = -3 \text{ cm}$ $f = 4 \text{ cm}$ (I) Using $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ $\frac{1}{v} = \frac{1}{u} + \frac{1}{f} = \frac{1}{-3.0} + \frac{1}{4.0}$ $\frac{1}{v} = \frac{-4+3}{12} = -\frac{1}{12} \quad v = -12 \text{ cm}$ (II) Linear magnification $m = \frac{v}{u}$ $m = \frac{-12}{-3} = 4$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ 1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	3
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	(c) As $\delta = \delta_m$ $r_1 = r_2$ (Alternatively) Hence, r_1 (Angle of refraction at P) $1 \times \sin 45 = \mu \sin r_1$ $1 \times \frac{1}{\sqrt{2}} = \sqrt{2} \sin r_1$ $\therefore \angle r_1 = 30^\circ$	$\frac{1}{2}$ $\frac{1}{2}$	3								
11.	a) <table border="1"> <tr> <td>• Identification of e.m. wave</td> <td>$1\frac{1}{2}$</td> </tr> <tr> <td>• Frequency range</td> <td>$1\frac{1}{2}$</td> </tr> </table> (i) Microwave $10^{10}\text{Hz} - 10^{12}\text{Hz}$ (ii) X-rays. $10^{16}\text{Hz} - 10^{20}\text{Hz}$ Alternatively Gamma Ray $10^{20}\text{Hz} - 10^{24}\text{Hz}$ (iii) Gamma Ray $10^{20}\text{Hz} - 10^{24}\text{Hz}$ Alternatively Infrared Radiations $10^{12}\text{Hz} - 10^{14}\text{Hz}$ (b) OR <table border="1"> <tr> <td>• Showing refracted wavefront</td> <td>1</td> </tr> <tr> <td>• Verification of Snell's law</td> <td>2</td> </tr> </table>	• Identification of e.m. wave	$1\frac{1}{2}$	• Frequency range	$1\frac{1}{2}$	• Showing refracted wavefront	1	• Verification of Snell's law	2	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	1
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	We consider, refracted wavefront CE and triangles ABC & AEC. From the triangles we obtained $\sin i = \frac{BC}{AC} = \frac{v_1 \tau}{AC}$ $\sin r = \frac{AE}{AC} = \frac{v_2 \tau}{AC}$ Thus $\frac{\sin i}{\sin r} = \frac{v_2}{v_1}$ We know $n = \frac{c}{v}$, So $\frac{\sin i}{\sin r} = \frac{n_2}{n_1}$ which is the Snell's law.	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	3
12.	(a) (iv) $\frac{n_1}{v} - \frac{n_2}{u} = \frac{(n_1 - n_2)}{R}$ (b) (iv) virtual and formed in air (c) (i) real and of the size of the object (d) (ii) + 5D (e) (iii) f	1 1 1 1 1	5

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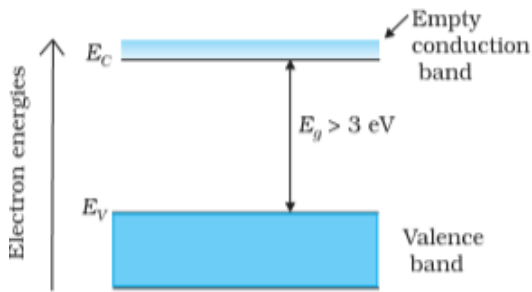
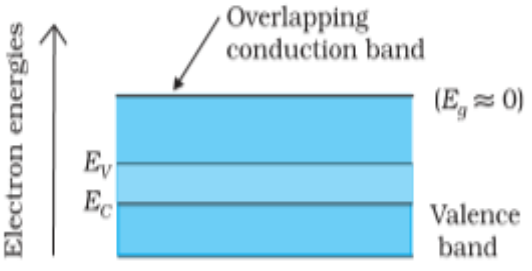
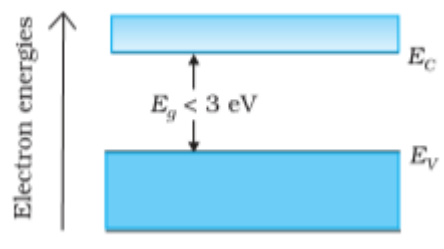
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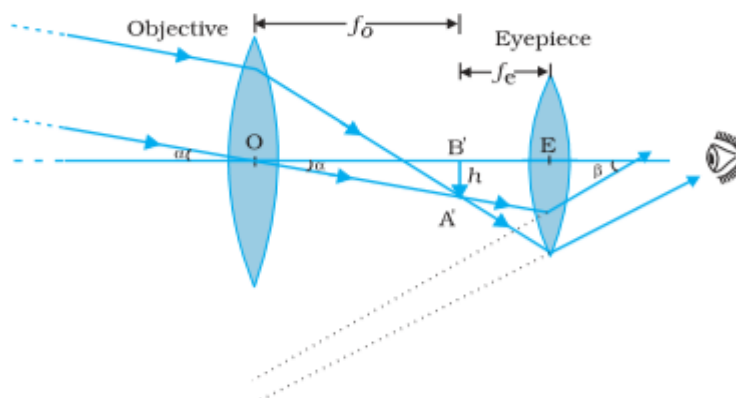
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2.	- Naming the device 1 - Principle of working 1 ❖ Rectifier ❖ The unidirectional property of a diode makes it suitable for reflection. Alternatively: The diode conducts when forward biased and does not conduct when reverse biased	1 1	2
3.	a) - Name of Spectral Series $\frac{1}{2}$ - Ratio of the wavelength $1\frac{1}{2}$ Balmer Series $\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$ $\frac{1}{\lambda_{\max}} = R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) \quad (n_f = 2, \quad n_i = 3)$ $\lambda_{\max} = \frac{36}{5R}$ $\frac{1}{\lambda_{\min}} = R \left(\frac{1}{2^2} - \frac{1}{\infty} \right) \quad (n_f = 2, \quad n_i = \infty)$ $\lambda_{\min} = \frac{4}{R}$ $\frac{\lambda_{\max}}{\lambda_{\min}} = \frac{9}{5}$ OR b) - Meaning of matter waves $\frac{1}{2}$ - Ratio of de- Broglie wavelength $1\frac{1}{2}$ Wave associated with moving material particle $\lambda = \frac{h}{\sqrt{2mqmV}}$ $\therefore \frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{q_\alpha m_\alpha}{q_p m_p}}$ $\therefore \frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{2e \cdot 4m}{e \cdot m}}$ $= \frac{2\sqrt{2}}{1}$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	2

5.	a) Calculation of distance of closest approach for α- particles $1\frac{1}{2}$ b) Calculation of distance of closest approach for photon $1\frac{1}{2}$ a) For α particles, distance of closest approach $r_{\alpha} = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{E_k}$ $r_{\alpha} = \frac{9 \times 10^9 \times 2 \times 79 \times (1.6 \times 10^{-19})^2}{2.56 \times 10^{-12}}$ $= 14.22 \times 10^{-15} \text{ m}$ $= 14.22 \text{ fm}$ b) For proton, distance of closest approach $r_p = \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{E_k}$ $r_p = \frac{r_{\alpha}}{2}$ $= 7.11 \times 10^{-15} \text{ m}$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	3
6.	a) For explaining reason 1 b) For explaining reason 1 c) For explaining reason 1 a) When a p-n junction is forward biased, the junction width decreases and as a result, its resistance also decreases. On the other hand, when a p-n junction is reverse biased, the junction width increases and as a result its resistance also increases. b) Conductivity of intrinsic semi-conductors is very low. Hence, no electronic device can be developed using them. Dopping increases conductivity, hence makes intrinsic semiconductor suitable for making electronic devices. c) It is easier to observe the change in the current with change in light intensity if a reverse bias is applied. Alternatively The fractional change due to photo-effects on the minority charge carriers dominated reverse bias current, is more easily measurable than the fractional change in forward bias current.	$\frac{1}{2}$ $\frac{1}{2}$ 1 1	3

7.	a) Explanation1 b) Condition for coherent source1 c) Explanation1 a) $\beta = \frac{\lambda D}{d}$, When space between point source increases, fringe width decrease. Interference pattern gets less and less sharp/ Interference pattern will not be observable. Alternatively On increasing separation between the slits the condition $\frac{s}{S} < \frac{\lambda}{d}$, will not be satisfied and interference pattern disappears. b) Two sources must have same frequency and constant or zero phase difference / constant or zero path difference. c) $\beta \propto \lambda$ fringe width will increase & intensity (Brightness) of fringe will decrease.	1 <
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(i)



1

(ii) Given $f_o = 150 \text{ cm}$, $f_e = 6 \text{ cm}$

(I) Length of the tube $L = f_o + f_e$
 $= 150 + 6$

$\frac{1}{2}$

$$L = 156 \text{ cm}$$

$\frac{1}{2}$

(II) $m = \frac{f_o}{f_e}$

$\frac{1}{2}$

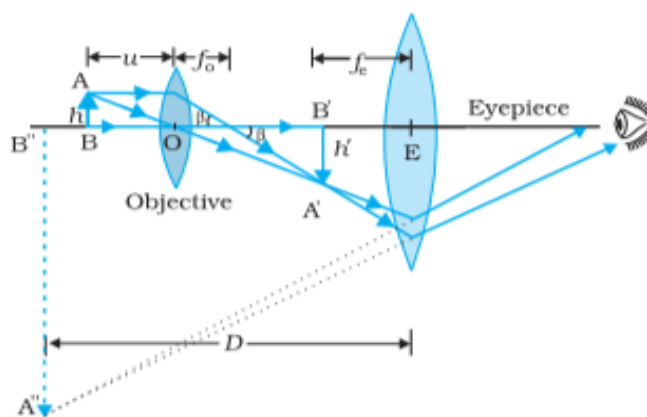
$$\frac{150}{6} = 25$$

$\frac{1}{2}$

OR

- b) (i) Labelled ray diagram of compound microscope–
(ii) (I) Position of image calculation 1
(II) Calculation of linear magnification 1

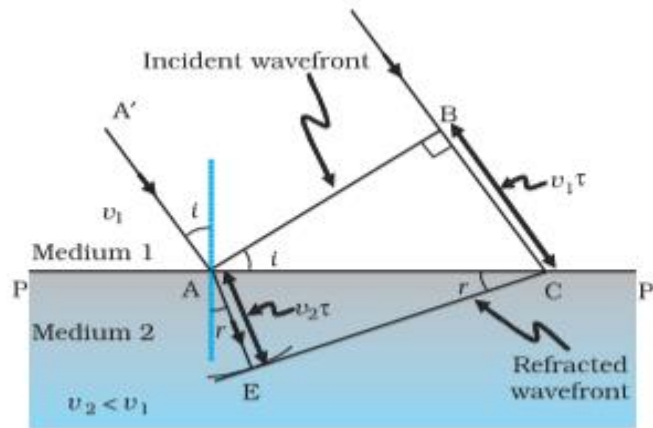
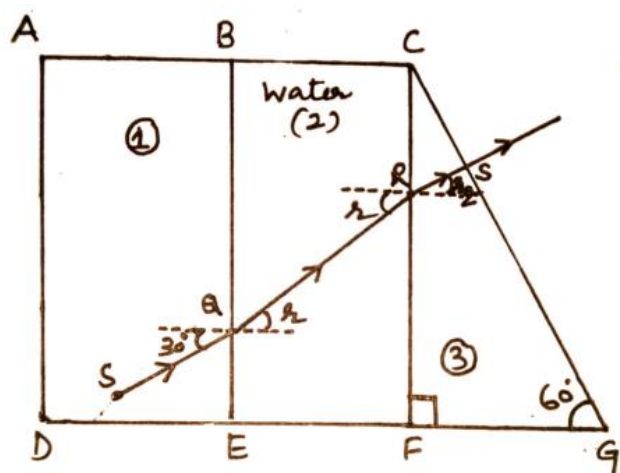
(i)



1

Ray diagram of image formation by a compound microscope

	(ii) Given $u = -3 \text{ cm}$ $f = 4 \text{ cm}$ (I) Using $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ $\frac{1}{v} = \frac{1}{u} + \frac{1}{f} = \frac{1}{-30} + \frac{1}{40}$ $\frac{1}{v} = \frac{-4+3}{12} = -\frac{1}{12} \quad v = -12 \text{ cm}$ (II) Linear magnification $m = \frac{v}{u}$ $m = \frac{-12}{-3} = 4$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	3												
10.	a) <table border="1"> <tr> <td>(i)</td> <td>Identification of e.m. wave</td> <td>$1\frac{1}{2}$</td> </tr> <tr> <td>(ii)</td> <td>Frequency range</td> <td>$1\frac{1}{2}$</td> </tr> </table> (i) Microwave $10^{10} \text{ Hz} - 10^{12} \text{ Hz}$ (ii) X-rays. $10^{16} \text{ Hz} - 10^{20} \text{ Hz}$ Alternatively Gamma Ray $10^{20} \text{ Hz} - 10^{24} \text{ Hz}$ (iii) Gamma Ray $10^{20} \text{ Hz} - 10^{24} \text{ Hz}$ Alternatively Infrared Radiations $10^{12} \text{ Hz} - 10^{14} \text{ Hz}$ OR b) <table border="1"> <tr> <td>•</td> <td>Showing refracted wavefront</td> <td>1</td> </tr> <tr> <td>•</td> <td>Verification of Snell's law</td> <td>2</td> </tr> </table>	(i)	Identification of e.m. wave	$1\frac{1}{2}$	(ii)	Frequency range	$1\frac{1}{2}$	•	Showing refracted wavefront	1	•	Verification of Snell's law	2	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	
(i)	Identification of e.m. wave	$1\frac{1}{2}$													
(ii)	Frequency range	$1\frac{1}{2}$													
•	Showing refracted wavefront	1													
•	Verification of Snell's law	2													

	 We consider, refracted wavefront CE and triangles ABC & AEC. From the triangles we obtained $\sin i = \frac{BC}{AC} = \frac{v_1 \tau}{AC}$ $\sin r = \frac{AE}{AC} = \frac{v_2 \tau}{AC}$ Thus $\frac{\sin i}{\sin r} = \frac{v_2}{v_1}$ We know $n = \frac{c}{v}$, So $\frac{\sin i}{\sin r} = \frac{n_2}{n_1}$ which is the Snell's law.	1	
11.	a) Tracing the path of emerging ray1 b) Finding the angle of emergence2 a) 		3

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SeniorSecondary School ,Term II Examination2022

Marking Scheme – PHYSICS (SUBJECT CODE – 042)

(PAPER CODE – 55/4/3)

General Instructions: -

1. You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully.
2. **“Evaluation policy is a confidential policy as it is related to the confidentiality of the examinations conducted, Evaluation done and several other aspects. Its’ leakage to public in any manner could lead to derailment of the examination system and affect the life and future of millions of candidates. Sharing this policy/document to anyone, publishing in any magazine and printing in News Paper/Website etc may invite action under IPC.”**
3. Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one’s own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. **However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and marks be awarded to them. In class-X, while evaluating two competency based questions, please try to understand given answer and even if reply is not from marking scheme but correct competency is enumerated by the candidate, marks should be awarded.**
4. The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
5. Evaluators will mark($\checkmark$) wherever answer is correct. For wrong answer ‘X’ be marked. Evaluators will not put right kind of mark while evaluating which gives an impression that answer is correct and no marks are awarded. **This is most common mistake which evaluators are committing.**
6. If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled. This may be followed strictly.
7. If a question does not have any parts, marks must be awarded in the left-hand margin and encircled. This may also be followed strictly.
8. If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out.
9. No marks to be deducted for the cumulative effect of an error. It should be penalized only once.

10. A full scale of marks 0-35 has to be used. Please do not hesitate to award full marks if the answer deserves it.
11. Every examiner has to necessarily do evaluation work for full working hours i.e. 8 hours every day and evaluate 30 answer books per day in main subjects and 35 answer books per day in other subjects (Details are given in Spot Guidelines). This is in view of the reduced syllabus and number of questions in question paper
12. Ensure that you do not make the following common types of errors committed by the Examiner in the past:-
 - Leaving answer or part thereof unassessed in an answer book.
 - Giving more marks for an answer than assigned to it.
 - Wrong totaling of marks awarded on a reply.
 - Wrong transfer of marks from the inside pages of the answer book to the title page.
 - Wrong question wise totaling on the title page.
 - Wrong totaling of marks of the two columns on the title page.
 - Wrong grand total.
 - Marks in words and figures not tallying.
 - Wrong transfer of marks from the answer book to online award list.
 - Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.)
 - Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
13. While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as cross (X) and awarded zero (0) Marks.
14. Any unassessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
15. The Examiners should acquaint themselves with the guidelines given in the Guidelines for spot Evaluation before starting the actual evaluation.
16. Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.
17. The Board permits candidates to obtain photocopy of the Answer Book on request in an RTI application and also separately as a part of the re-evaluation process on payment of the processing charges.

MARKING SCHEME
 Senior Secondary School Examination TERM–II, 2022
PHYSICS (Subject Code–042)
[Paper Code : 55/4/3]

Q. No.	EXPECTED ANSWER / VALUE POINTS	Marks	Total Marks
1.	a) - Name of Spectral Series $\frac{1}{2}$ - Ratio of the wavelength $1\frac{1}{2}$ Balmer Series $\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$ $\frac{1}{\lambda_{\max}} = R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) \quad (n_f = 2, \quad n_i = 3)$ $\lambda_{\max} = \frac{36}{5R}$ $\frac{1}{\lambda_{\min}} = R \left(\frac{1}{2^2} - \frac{1}{\infty} \right) \quad (n_f = 2, \quad n_i = \infty)$ $\lambda_{\min} = \frac{4}{R}$ $\frac{\lambda_{\max}}{\lambda_{\min}} = \frac{9}{5}$ OR b) - Definition /meaning of matter waves $\frac{1}{2}$ - Ratio of de- Broglie Wavelength $1\frac{1}{2}$ Wave associated with moving material particle $\lambda = \frac{h}{\sqrt{2qmV}}$ $\therefore \frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{q_\alpha m_\alpha}{q_p m_p}}$ $\therefore \frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{2e \cdot 4m}{e \cdot m}}$ $= \frac{2\sqrt{2}}{1}$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	2

2.	• Meaning of energy Band gap 1/2 • Energy Band Diagram of Conductor, Insulator & Semiconductor (1/2 + 1/2 + 1/2) Energy gap : The gap between the top of the valence band and bottom of the conduction band is called the energy band gap. (b) ENERGY BAND DIAGRAM FOR INSULATOR ENERGY BAND DIAGRAM FOR CONDUCTOR ENERGY BAND DIAGRAM FOR SEMICONDUCTOR	1/2 1/2 1/2	2
3.	• Definition of Barrier Potential 1 • Explanation of thickness of depletion region 1 (i) It is the potential developed across the p-n junction which tends to prevent the movement of majority charge carriers from both sides. (ii) Under reverse bias, direction of applied voltage is same as the direction of barrier potential. As a result barrier height increase & the depletion region widens due to change in electric field.	1 1	2

	SECTION-B		
4.	a) For explaining reason 1 b) For explaining reason 1 c) For explaining reason 1 a) When a $p-n$ junction is forward biased, the junction width decreases and as a result, its resistance also decreases. On the other hand, when a $p-n$ junction is reverse biased, the junction width increases and as a result, its resistance also increases. b) Conductivity of intrinsic semi-conductors is very low. Hence, no electronic device can be developed using them. Dopping increases conductivity, hence makes intrinsic semiconductor suitable for making electronic devices. c) It is easier to observe the change in the current with change in light intensity if a reverse bias is applied. Alternatively The fractional change due to photo-effects on the minority charge carriers dominated reverse bias current, is more easily measurable than the fractional change in forward bias current.	$\frac{1}{2}$ $\frac{1}{2}$ 1 1	3

6.

Brief Explanation of bright and dark fringes in diffraction

2

Explanation of decrease in intensity with increasing 'n'

1

Dividing the slit into smaller parts, and adding their contributions at P with the proper phase differences. The path difference $NP - LP$ between the two edges of the slit, $NP - LP = NQ = a \sin \theta \approx a\theta$

$\frac{1}{2}$

At the central point C on the screen, the angle θ is zero. All path differences are zero and hence all the parts of the slit contribute in phase. This gives maximum intensity at C,

$\frac{1}{2}$

Similarly other secondary maxima will be formed at $\theta = (n+1/2) \lambda/a$,

For minima (zero intensity) the contributions from M_1 and M_2 are 180° out of phase and cancel in the direction $\theta = \lambda/a$. where λ is the path difference. Similarly other minima are formed due to cancellation of contributions. At an angle $\theta = n\lambda/a$, where $n = \pm 1, \pm 2, \pm 3, \dots$

$\frac{1}{2}$

$\frac{1}{2}$

Consider an angle $\theta = 3\lambda/2a$ which is midway between two of the dark fringes. Divide the slit into three equal parts. If we take the first two thirds of the slit, the path difference between the two ends would be λ .

$\frac{1}{2}$

The first two-thirds of the slit can be divided into two halves which have a $\lambda/2$ path difference. The contributions of these two halves cancel, only the remaining one-third of the slit contributes to the intensity at a point between the two minima. Which will be much weaker than the central maximum (where the entire slit contributes in phase). Similarly there are maxima at $(n + 1/2) \lambda/a$ with $n = 2, 3$, etc. These become weaker with increasing n , since only one-fifth, one-seventh, etc., of the slit contributes in these cases.

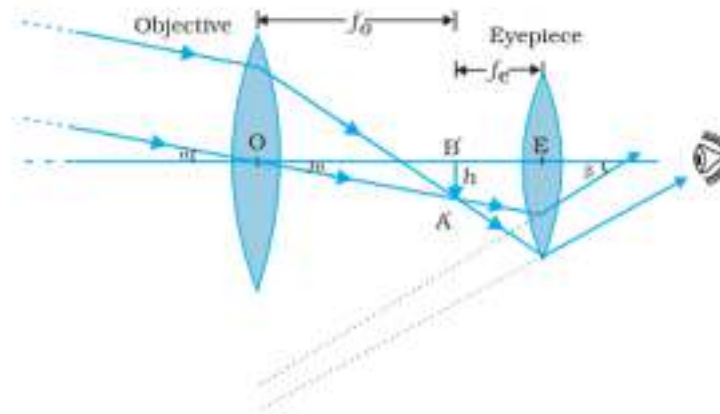
$\frac{1}{2}$

3

7.

- | | |
|-------------------------------------|---|
| a) (i) Labelled Diagram – | 1 |
| (ii) (I) Length of tube calculation | 1 |
| (II) Calculation of magnification | 1 |

(a) (i)



(ii) Given $f_o = 150 \text{ cm}$, $f_e = 6 \text{ cm}$

(I) Length of the tube $L = f_o + f_e$

$$= 150 + 6$$

$$L = 156 \text{ cm}$$

(II) $m = \frac{f_o}{f_e}$

$$\frac{150}{6} = 25$$

OR

- | | |
|--|---|
| b) (i) Labelled Diagram – | 1 |
| (ii) (I) Position of image calculation | 1 |
| (II) Calculation of linear magnification | 1 |

1

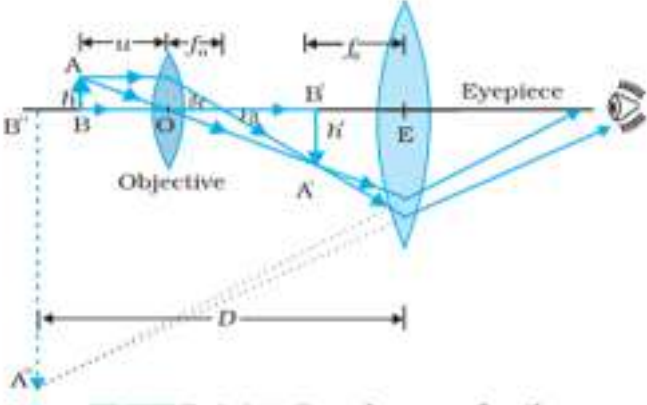
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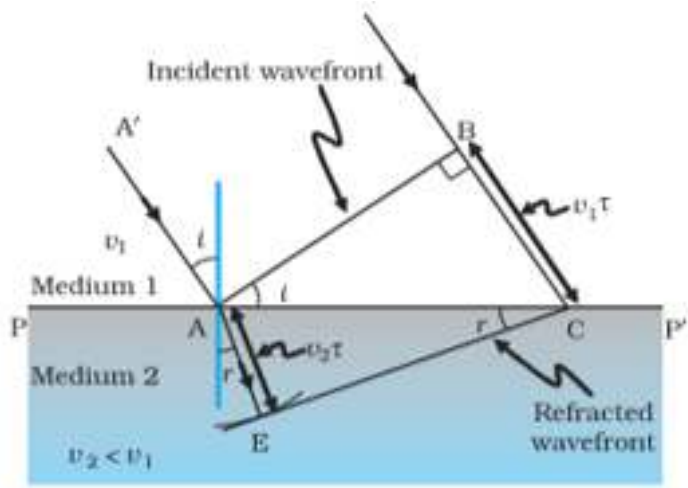
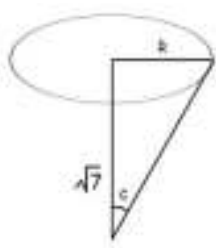
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

3

	(b) (i)  Ray diagram of image formation by a compound microscope (ii) Given $u = -3 \text{ cm}$ $f = 4 \text{ cm}$ (I) Using $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ $\frac{1}{v} = \frac{1}{u} + \frac{1}{f} = \frac{1}{-3.0} + \frac{1}{4.0}$ $\frac{1}{v} = \frac{-4 + 3}{12} = -\frac{1}{12} \quad v = -12 \text{ cm}$ (II) Linear magnification $m = \frac{v}{u}$ $m = \frac{-12}{-3} = 4$	1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	
8.	- Finding de- Broglie wavelength $1\frac{1}{2}$ - Finding the momentum of α particles $1\frac{1}{2}$ $\lambda_p = \frac{h}{\sqrt{2q_p m_p V}},$ $\lambda_\alpha = \frac{h}{\sqrt{2 \cdot 2q_p \cdot 4m_p V}}$ $\frac{\lambda_\alpha}{\lambda_p} = \frac{1}{2\sqrt{2}}$ $\lambda_\alpha = \frac{h}{p_p}, \quad \lambda_\alpha = \frac{h}{p_\alpha}$ $\frac{p_\alpha}{p_p} = \frac{1}{2\sqrt{2}},$ $p_\alpha = 2\sqrt{2} p_p$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	3

	 We consider, refracted wavefront CE and triangles ABC & AEC. From the triangles we obtained $\sin i = \frac{BC}{AC} = \frac{v_1 \tau}{AC}$ $\sin r = \frac{AE}{AC} = \frac{v_2 \tau}{AC}$ Thus $\frac{\sin i}{\sin r} = \frac{v_2}{v_1}$ We know $n = \frac{c}{v}$, So $\frac{\sin i}{\sin r} = \frac{n_2}{n_1}$ which is the Snell's law.	1	
11.	• Calculation of critical angle 1 • Calculation of radius circle $1\frac{1}{2}$ • Calculation of Area $\frac{1}{2}$  $\sin C = \frac{1}{\mu}$ $\sin C = \frac{3}{4}$ $\sin C = \frac{R}{\sqrt{R^2 + 7}}$		

	$\frac{3}{4} = \frac{R}{\sqrt{R^2 + 7}}$ $R = 3\text{m}$ $\text{Area} = \pi R^2$ $= 3.14 \times (3)^2$ $= 28.26 \text{ m}^2$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	3
12.	(a) (iv) $\frac{n_1}{v} - \frac{n_2}{u} = \frac{(n_1 - n_2)}{R}$ (b) (iv) virtual and formed in air (c) (i) real and of the size of the object (d) (ii) +5D (e) (iii) f	1 1 1 1 1	5

* * *